

Supplementary S1 Complete text of the online survey instrument

Note: The online survey instrument is no longer available, as Google Forms removes surveys after they've been closed. However, full survey text, as it appeared online, is included herein.

Survey Title: What types of information provided by wildlife scientists are most relevant and/or useful to forest managers in eastern North America?

Purpose: The goal of this survey is to get feedback from forest management practitioners on which types of information from wildlife scientists would be most directly relevant and/or useful to managers that need to: 1) describe baseline wildlife habitat conditions, 2) specify desired future condition targets for wildlife, 3) evaluate progress toward meeting wildlife objectives, and/or 4) evaluate constraints that wildlife habitat may impose on other forest management objectives.

How this information will be used: Survey results will be used to develop relevance criteria that we will apply to classify different articles on forest bird habitat relationships in a review paper titled: "Do review papers on forest bird-vegetation relationships provide actionable information to forest managers in eastern North America?" that will be submitted for a special issue of the journal *Forests* titled: "Evidence synthesis in forest management". The full list of review team members with their affiliations is included at the end of this survey.

Confidentiality: All surveys will be stripped of respondent names and contact information prior to analysis. The contact information requested below will be used to ask follow-up questions, if necessary. All participants may specify below that they would prefer not to be contacted with follow up questions. We may summarize information on affiliation types and professional roles of survey respondents in our publication, but no individual or organizational names will be included.

How long will this survey take me to complete? The survey is comprised of multiple-choice questions on 6 different topics. Each topic has an optional open-ended section where managers can identify specific types of information that they would like to receive from wildlife scientists. Completing the survey should take 5-15 minutes, depending on how much time individual respondents spend on open ended questions.

When and where should I return my completed survey? Please return all completed surveys by email to Casey Lott at caseylott@ymail.com prior to December 15th. You may contact Casey by email or phone at (208) 629-8705 if you have any questions.

RESPONDENT INFORMATION

What is your affiliation type? Select only one.

- ☐ Federal agency
- ☐ State agency
- ☐ Industrial timber company
- ☐ Consulting Forester
- ☐ Cooperative Extension Forester
- ☐ Private NGO
- ☐ Other: _____

What is your professional role? Select all that apply.

- ☐ Forester
- ☐ Forest planner
- ☐ Wildlife scientist
- ☐ Other: _____

Which landscape sizes do you work in? Select all that apply.

- ☐ Less than 1,000 acres
- ☐ 1,000 to 10,000 acres
- ☐ 10,000 to 100,000 acres
- ☐ 100,000 to 500,000 acres
- ☐ 500,000 to 1,000,000 acres
- ☐ More than 1,000,000 acres

Which forest regions do you work in? Select all that apply.

- ☐ Beech/Maple/Basswood
- ☐ Mississippi Alluvial Plain
- ☐ Mixed Mesophytic
- ☐ Northern Hardwood-Hemlock
- ☐ Northern Hardwood-Red Pine
- ☐ Oak-Hickory
- ☐ Southern Mixed
- ☐ Subtropical Evergreen

SURVEY QUESTIONS

For each question, select all check boxes that apply. Use the "other" line to write in answers that are not provided as check box options. After filling in boxes for each question, please describe any specific information needs that you have from wildlife scientists in the space provided.

1. At which spatial scales do you use forest attribute data? Select all that apply.

- ☐ Individual trees
- ☐ Plots or transects within a stand
- ☐ Whole stand metrics
- ☐ Management zones/compartments within a defined forest
- ☐ Whole forest
- ☐ Landscapes larger than a single forest
- ☐ Region

☐ Other: _____

Q: Given the spatial scales that are important to you as a forest manager, which kinds of information would you like wildlife scientists to provide at each of these scales? Please list any specific information you think wildlife scientists should provide so that data collected during wildlife studies can be linked to the forest attributes that are most relevant to you as a forest manager at each of your operational spatial scales. Describe below...

2. At which temporal scales do you need forest attribute data? Select all that apply.

- ☐ Dates associated with specific events
- ☐ Intensive data from a single sampling period
- ☐ Annual data collected within pre-defined seasonal time periods (e.g., spring surveys)
- ☐ A time series of annual measurements over many years
- ☐ A time series of measurements at standard temporal increments (once every 3 years)
- ☐ Other: _____

Q: Given the temporal scales that are important to you as a forest manager, which kinds of information would you like wildlife scientists to provide at each of these scales? Please list any specific information that you think wildlife scientists should provide so that data collected during wildlife studies can be linked to the forest attributes that are most relevant to you as a forest manager at each of your operational temporal scales. Describe below...

3. Which within-stand elements do you use forest attribute data for? Select all that apply.

- ☐ Forest litter
- ☐ Herb-forb percent cover or density
- ☐ Shrub/understory percent cover or density
- ☐ Sapling-pole/mid-story percent cover or density
- ☐ Canopy percent cover or density
- ☐ Herb-forb species composition
- ☐ Shrub/understory species composition
- ☐ Sapling-pole/mid-story species composition
- ☐ Canopy species composition
- ☐ Canopy gaps
- ☐ Tree height
- ☐ Tree age
- ☐ Tree size/dbh
- ☐ Tree basal area

- ☐ Tree density
- ☐ Live cavity trees
- ☐ Snags
- ☐ Logs/coarse woody debris
- ☐ Slash
- ☐ Water features
- ☐ Bare ground
- ☐ Soft mast
- ☐ Hard mast
- ☐ Other: _____

Q: Given the types of within-stand information that are important to you as a forest manager, which kinds of within-stand information would you like wildlife scientists to provide? Please list any specific information that you think wildlife scientists should measure so that data collected during wildlife studies can be linked to the types of data on within-stand attributes that are most relevant to you as a forest manager. Describe below...

4. Which stand-level variables (that are not based on summaries of plot data) do you need information for? Select all that apply.

- ☐ Forest type
- ☐ Stand Age
- ☐ Age class
- ☐ Size class
- ☐ Timber class
- ☐ Stand Size
- ☐ Stand Origin
- ☐ Stocking
- ☐ Site index
- ☐ Site class
- ☐ Management history
- ☐ Other: _____

Q: Given the types of stand-level information that are important to you as a forest manager, which kinds of stand-level information would you like wildlife scientists to provide? Please list any specific information that you think wildlife scientists should provide so that data collected during wildlife studies can be linked to the types of data on stand-level attributes that are most relevant to you as a forest manager. Describe below...

5. Which forest, landscape, or region scale variables do you need information for? Select all that apply.

- ☐ Forested area
- ☐ Forested area by forest type
- ☐ Forested area by age class
- ☐ Forested area by size class
- ☐ Forested area by timber class
- ☐ Non-forested area by land cover type
- ☐ Number of stands of a certain type
- ☐ Stand size distributions
- ☐ Edge-related statistics
- ☐ Core-area related statistics
- ☐ Forested area by management zone type
- ☐ Other: _____

Q: Given the types of forest, landscape, or region- scale information that are important to you as a forest manager, which kinds of forest, landscape, or region scale information would you like wildlife scientists to provide? Please list any specific information that you think wildlife scientists should provide so that data collected during wildlife studies can be linked to the forest, landscape, or region scale attributes that are most relevant to you as a forest manager. Describe below...

6. For which of the following applications do you need information from wildlife scientists? Select all that apply.

- ☐ Providing general guidance on the value of different wildlife habitat objectives
- ☐ Describing wildlife habitat as a set of desired forest attributes at plot or stand scales
- ☐ Describing the amount of existing wildlife habitat at forest, landscape, or regional scales
- ☐ Estimating the amount of wildlife habitat that is needed for a given species at forest, landscape, or regional scales
- ☐ Documenting changes in the amount of wildlife habitat at forest, landscape, or regional scales between two or more time periods
- ☐ Describing connectivity requirements for habitat objectives
- ☐ Describing proximity requirements for habitat objectives
- ☐ Describing wildlife habitat-related constraints on meeting other non-wildlife forest management objectives.
- ☐ Other: _____

Q: Given the types of applied forest management activities that are important to you as a forest manager, which kinds of applied information would you like wildlife scientists to provide? Please list

any specific information that you think wildlife scientists should provide so that data collected during wildlife studies can be linked to the forest management applications that are most relevant to you as a forest manager.

Q: What are your methods for stand developmental stage classification? Please describe below...

Thank you for taking the time to respond to this survey!

Review team: **Casey A. Lott**, Conservation Science Institute, Boise, ID, USA and Indiana University of Pennsylvania-Research Institute, Indiana, PA, USA; **Michael E. Akresh**, Environmental Studies Department, Antioch University New England ; **Cameron J. Fiss**, Department of Environmental and Forest Biology, State University of New York College of Environmental Science and Forestry, Syracuse, NY, USA.; **David I. King**, USDA Forest Service, Northern Research Station, University of Massachusetts, Amherst, MA, USA.; **Scott H. Stoleson**, USDA Forest Service, Northern Research Station, Irvine, PA, USA.; **Jeffery L. Larkin**, Department of Biology, Indiana University of Pennsylvania, Indiana, PA, USA. and American Bird Conservancy, The Plains, VA, USA.

Note: This survey instrument was reviewed by the Institutional Review Board at Antioch University prior to distribution.